

Student Expense Tracker

Overview

Student Expense Tracker is a small Python application that I made to note down my daily expenses and see how much I am spending in a month. It has a simple window (made with Tkinter) where we can add an expense, search it, delete it, set a monthly budget and see a report. All the data is saved in a JSON file on the computer, so nothing is lost when the program is closed.

This is my first-semester Python project. While making it I used the things we learned in class like functions, classes, lists, dictionaries, loops, conditions, file handling, exception handling and basic GUI programming.

Problem Statement

Students spend money many times in a day, for example on food, travel, stationery, books, snacks and online shopping. Most of us do not write these things down, so at the end of the month we do not know where the money went or whether we stayed inside our budget or we have overspent this month. This project gives a simple way to record all these expenses at one place and check them later.

Objectives

- Record daily expenses.
- Keep expenses in categories.
- Search and filter the saved expenses.
- Delete records that are not needed.
- Find the highest expense.
- Calculate how much is spent in the current month.
- Set a monthly budget and keep track of it.
- Generate a simple expense report.
- Save and load the data locally.

Features

- Add an expense with amount, category, description and date (the date box already shows today's date, and if it is left empty today's date is used).
- Shows all the expenses in a table.
- Filter the table by category (there are 8 categories: Food, Travel, Stationery, Books, Snacks, Necessities, Online Shopping and Entertainment). Search using text (it checks the description and the category).
- Delete the selected expense (it asks for confirmation first).
- Show the highest expense.

- • Set a monthly budget.
- Shows the money spent this month and the budget status in green, orange, red or gray colour.
- • A warning box comes when 80% of the budget is used or when the budget is crossed.
- Generate an expense report (total, average, highest, lowest and category-wise spending).
- Saves everything in expenses.json .

Technologies

- Python 3
- Tkinter (for the GUI)
- JSON (for saving data)
- Git/GitHub
- VS Code

Matplotlib is not needed in the final version. Only the built-in Python modules are used, so nothing extra has to be installed.

Project Structure

```
expense_tracker
├── main.py
├── gui.py
└── expense_manager.py
```

└─ budget.py
└─ reports.py
└─ storage.py
└─ expenses.json
└─ README.md
└─ statement.md

└─ tests.py # not added yet,

I want to add it before final submission

expenses.json is created by the program when the
first expense or budget is saved.

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File Description

- main.py - starts the application.
- gui.py - has the Tkinter window, the buttons and what happens when they are clicked.
- expense_manager.py - has the ExpenseManager class which adds, deletes, searches and calculates expenses.
- budget.py - checks the budget entered by the user and tells the budget status.
- reports.py makes the expense report and the spending totals.
- storage.py saves the data to the JSON file and loads it back.
- expenses.json the saved expenses and budget

• **Installation and Running**

1. Install Python 3 on your computer (Tkinter comes
 - along with it in most cases).
2. Open the project folder in VS Code.
3. Run this command in the terminal:

```
python3 main.py
```

On some computers the command may be.
pythonmain.py instead. No external library is needed.

Testing

Right now I have tested the project manually. The steps are:

1. Add a valid expense.
2. Try an invalid amount or a wrong date and check the error message.
3. Filter by category.
4. Search for an expense.
5. Delete a selected expense.
6. Set a monthly budget.
7. Check the budget warning/status.
8. Open the expense report.
9. Close the program, open it again and check that the saved data is loaded.

I have not written a tests.py file yet. I think it is a good idea to add one for automatic testing before the final submission

Functional Requirements

1. Add expenses.
2. Display and delete expenses.
3. Search and filter expenses.
4. Calculate monthly spending.
5. Set a monthly budget.
6. Monitor the budget status.
7. Generate reports.
8. Save and load data.

Non-Functional Requirements

- **Usability:** the interface is simple, with forms, buttons and a table, so user can use it easily.
- **Reliability:** saved data is loaded again when the program starts, and wrong inputs are handled.
Maintainability: each file does one job, so it is easier to understand and change the code.
Performance: it is fine for the normal number of expenses a student will have.
- **Resource efficiency:** it only uses one small local JSON file.

- **Error handling:** if the input is wrong, a clear message is shown to the user.
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GitHub

The repository link will be in this format:

https://github.com/Vidushi022/expense_tracker

Future Enhancements

- Add automated unit tests.
- Export the data to a CSV file.
- Let the user add custom categories.
- A calendar view for expenses.
- Different profiles for multiple users.
- Charts, if external libraries are allowed.